

NTI – Data Analysis Final Project

Team Members

Nour Ayman Mosad
Habiba Mustafa ibrahim
Jana Ezzat Abdelhamid

Data Overview

This project presents a comprehensive analysis of an **e-commerce sales dataset containing 34,500 transaction records**, with the objective of transforming raw data into meaningful business insights related to **sales, customer behavior, product performance, profitability, returns, delivery efficiency, and sales trends**.

The analysis was conducted using **Python and Pandas**. Data preparation included inspecting the dataset, checking missing values and duplicates, validating data ranges and types, cleaning text fields, converting transaction dates to a standardized format, and creating a monthly feature for time-based analysis. The cleaned dataset was then used for further analysis and visualization.

Several key performance measures were calculated, including **total orders, total sales, average order value, total quantity sold, average profit margin, and total shipping cost**. The analysis further examined performance by **product category, region, payment method, return status, discount level, delivery time, and month**, providing insights into both financial and operational performance.

The findings are presented through **Power BI dashboards**, enabling interactive exploration of key performance indicators and trends. Overall, the project demonstrates an end-to-end **data analytics workflow**, from data preprocessing and analysis to business intelligence visualization, helping identify performance patterns and potential areas for improvement.

Data Cleaning & Transformation

Data Preprocessing and Feature Engineering

Before visualization and analysis in Power BI, the dataset was preprocessed and enhanced using **Python and the Pandas library** to ensure data quality, consistency, and analytical reliability.

- **Data Quality Checks:**

The dataset was checked for missing values and duplicate records using Pandas. Missing values were verified using `isnull()`, and no missing values were found across the 18 attributes. Duplicate rows were also checked using Pandas `duplicated()` function, with no duplicate records identified.

- **Data Type Standardization:**

Converted `order_date` into a proper datetime format and standardized financial attributes such as `price`, `total_amount`, `shipping_cost`, and `profit_margin` as numeric values to ensure accurate calculations and analysis.

- **Feature Engineering:**

Calculated `total_amount` based on the product price, applied discount, and quantity purchased. A month feature was extracted from `order_date` to support monthly and seasonal sales analysis. In addition, `customer_age` was grouped into meaningful age brackets to facilitate demographic and customer-segmentation analysis.

- **Outlier and Data Integrity Validation:**

Checked `delivery_time_days` to ensure that delivery durations remained within the expected operational range of **1–10 days**. Negative `profit_margin` values were also identified and analyzed as indicators of potentially loss-making transactions.

The preprocessing stage ensured that the dataset was **clean, consistent, and suitable for further analysis and visualization in Power BI**, providing a reliable foundation for the projects performance analysis and business insights.

Business Questions & Analysis

The analysis was designed to answer a set of **10 key business questions** related to sales performance, customer behavior, profitability, returns, discounts, and operational efficiency. These questions were selected to provide a comprehensive understanding of the e-commerce platform's performance and to support data-driven business decisions.

Core Key Performance Indicators (KPIs)

The main performance indicators identified from the dataset are:

- **Total Sales Revenue:** \$5,865,293.05
- **Total Net Profit:** \$970,019.41
- **Total Orders:** 34,500
- **Average Delivery Time:** 4.81 days
- **Overall Return Rate:** 5.52%

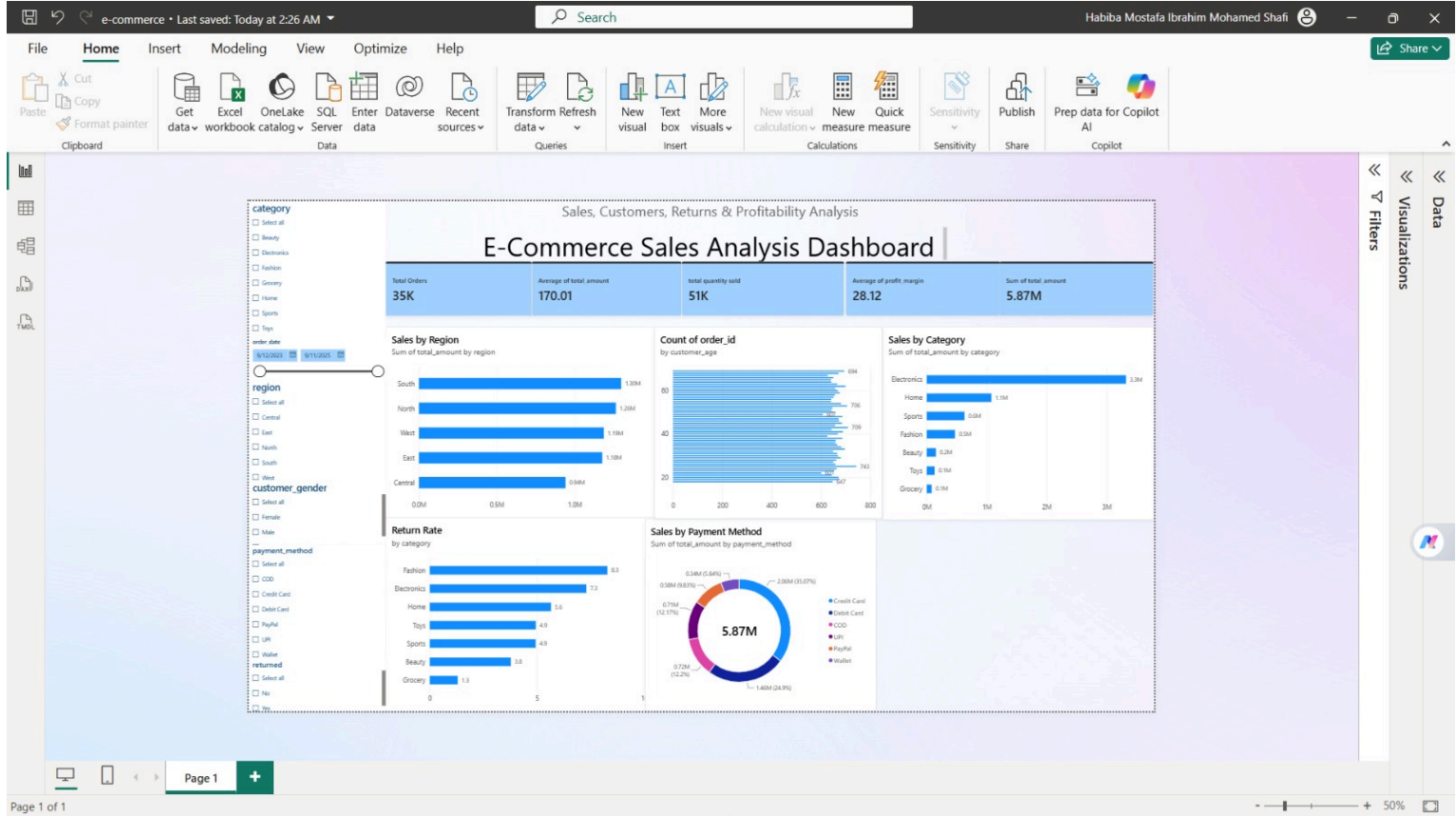
Key Business Questions

#	Business Question	Analysis Performed
1	Which product category generates the highest sales?	Sales by Category
2	Which region generates the highest sales?	Sales by Region
3	Which payment method is most popular among customers?	Payment Method Analysis
4	What is the overall order return rate?	Return Rate Analysis
5	Which product category has the highest return rate?	Returns by Category
6	How does the level of discount relate to sales performance?	Discount Analysis
7	Which product category sells the highest quantity of items?	Quantity by Category
8	Do returned orders have different delivery times compared with non-returned orders?	Delivery Time and Returns Analysis
9	Which product category generates the highest profit?	Profitability by Category
10	How do sales performance and order volume change over time?	Monthly Sales Analysis

Analytical Objective

These business questions guide the analysis of the e-commerce platform across **financial, operational, customer, and product areas**. The results are visualized in **Power BI dashboards** to identify key trends, performance indicators, and opportunities for improvement.

Dashboard



Key Findings & Recommendations

Key Findings:

1. **Electronics Dominance:** Electronics generated the highest total revenue (\$3.32M) and profit (\$344.3K), but showed a relatively high return rate of 7.30%.
2. **Unprofitable Grocery Segment:** The Grocery category generated \$82.0K in sales but recorded a **net profit loss of -\$9,187.96** due to high shipping costs relative to low item order values.
3. **Fashion Return Rates:** Fashion performed well in profit (\$128.8K) but suffered the highest return rate across all categories at **8.28%**.
4. **Balanced Regional Distribution:** Sales performance across regions was balanced, ranging from \$940.5K (Central) to \$1.30M (South).

Strategic Recommendations:

- **Revise Shipping Policy for Grocery Items:** Implement a minimum order threshold or adjust shipping cost formulas for low-margin items to eliminate net operational losses in the Grocery category.
- **Address High Return Rates in Fashion & Electronics:** Improve online sizing guides, product visualization, and quality checks for Fashion and Electronics to minimize return processing overhead.
- **Capitalize on High-Margin Categories:** Reallocate marketing budget toward **Electronics** and **Home** items, as they account for over 60% of total company gross profit.
- **Optimize Delivery Timelines:** Reduce average shipping times from 4.8 days down to under 3 days in key regions to improve customer satisfaction and reduce canceled/returned orders.

Conclusion

This project successfully transformed raw e-commerce transaction data into meaningful business insights to support data-driven decision-making. Python (Pandas) was used for data cleaning, preprocessing, and feature engineering, while Power BI was used to create interactive visualizations and dashboards.

The analysis provided valuable insights into sales performance, profitability, customer purchasing patterns, return behavior, and operational efficiency. The findings also highlighted key areas for improvement, particularly in managing unprofitable product categories, reducing high return rates, and improving delivery performance.

Overall, the project demonstrates how combining data preprocessing, analysis, and interactive business intelligence tools can help organizations better understand their performance and identify opportunities to improve profitability, customer satisfaction, and operational efficiency.